
1. Define robotics and explain its interdisciplinary nature

Robotics is a branch of engineering and science that deals with the **design, construction, programming, and application of robots** to perform tasks automatically or semi-automatically.

Interdisciplinary nature of robotics:

Robotics combines knowledge from multiple fields:

- **Mechanical Engineering** – robot structure, joints, actuators
- **Electrical & Electronics Engineering** – sensors, motors, circuits
- **Computer Science** – programming, algorithms, AI
- **Control Engineering** – motion control, stability
- **Artificial Intelligence** – learning, decision making
- **Mathematics** – kinematics, dynamics, optimization

☞ Hence, robotics is a **multidisciplinary field** integrating hardware and software.

2. Discuss the major historical milestones in the development of robotics

- **Ancient era:** Mechanical automata (Greek and Chinese inventions)
 - **1921:** Term “*Robot*” introduced by Karel Čapek
 - **1954:** First programmable robot (Unimate) by George Devol
 - **1961:** Unimate installed in General Motors factory
 - **1970s:** Industrial robots for welding and assembly
 - **1980s:** Introduction of sensors and microprocessors
 - **1990s:** Mobile robots and humanoids (Honda ASIMO)
 - **2000s–present:** AI-based robots, collaborative robots (cobots), autonomous robots
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3. What are the advantages and limitations of using exoskeletons for rehabilitation?

Advantages:

- Helps patients **regain movement and strength**
- Enables **early rehabilitation**
- Provides **repeatable and controlled therapy**
- Reduces physical strain on therapists

- Improves recovery outcomes

Limitations:

- High **cost**
- Limited adaptability to different body types
- Requires **technical supervision**
- Heavy or uncomfortable for long use
- Ethical and accessibility concerns

4. How has the field of robotics evolved from mechanical automation to AI-driven systems?

- Early robots followed **fixed mechanical instructions**
- Introduction of **sensors** enabled feedback control
- Microcontrollers allowed **programmable automation**
- Integration of **machine learning and AI**
- Robots can now:
 - Learn from experience
 - Recognize objects and speech
 - Make autonomous decisions
 - Collaborate safely with humans

☞ Robotics has evolved from **rule-based machines to intelligent systems**.

5. Compare and contrast first-generation and fourth-generation robots

Feature	First-Generation Robots	Fourth-Generation Robots
Intelligence	No intelligence	AI-based intelligence
Sensors	None	Advanced sensors
Adaptability	Fixed tasks	Adaptive & learning
Control	Open loop	Autonomous control
Examples	Early industrial robots	Self-driving robots, humanoids

6. Explain the significance of robotic autonomy levels and their applications

Levels of autonomy:

- **Manual control** – fully human-controlled
- **Semi-autonomous** – shared control
- **Fully autonomous** – independent decision making

Significance:

- Determines **human involvement**
- Enhances **safety and efficiency**
- Enables operation in **hazardous environments**

Applications:

- Autonomous vehicles
 - Space exploration
 - Medical robots
 - Military drones
 - Smart manufacturing
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7. Discuss the challenges of applying Asimov's Laws to autonomous systems and AI ethics

Challenges:

- Laws are **too abstract**
- Conflicts between laws are hard to resolve
- Robots lack moral understanding
- Context-based decisions are complex
- Responsibility and accountability issues

AI Ethics Issues:

- Bias in decision making
- Safety and transparency
- Privacy concerns
- Control over autonomous weapons

☞ Real-world systems need **practical ethical frameworks**, not fictional rules.

8. Explain how the DH parameter convention simplifies kinematic calculations

Denavit–Hartenberg (DH) convention is a standardized method to describe robot geometry.

Benefits:

- Reduces complex 3D transformations
- Uses only **four parameters per joint**
- Provides a systematic approach
- Simplifies forward kinematics
- Enables easy modeling of serial manipulators

☞ It makes robot motion analysis **consistent and manageable**.

9. How has industrial automation revolutionized manufacturing through robotics?

- Increased **production speed**
- Improved **accuracy and consistency**
- Reduced human error
- Enhanced worker safety
- Enabled 24/7 operation
- Reduced long-term costs
- Enabled mass customization

☞ Robotics has transformed manufacturing into **smart, flexible, and efficient systems**.

10. Discuss the concept of robot rights and whether robots should have legal status

Concept of robot rights:

Robot rights refer to granting **legal or moral status** to intelligent robots.

Arguments in favor:

- Advanced AI mimics human behavior
- Autonomous decision making
- Protection from misuse

Arguments against:

- Robots lack consciousness and emotions
- Humans create and control robots
- Legal responsibility must remain with humans

☞ Currently, robots are treated as **tools**, not legal entities. Legal responsibility lies with **manufacturers and users**.
